Going Public over the Business Cycle

Click here for the latest version

Jisu Jeun*

October 17, 2026

Abstract

This paper studies why initial public offering (IPO) activity is cyclical and how this cyclicality shapes aggregate fluctuations. Using data on U.S. nonfinancial firms, I document that IPO activity covaries with the business cycle along three margins. The number of IPOs and the amount of capital raised at listing are generally procyclical, whereas firms that go public during contractions subsequently grow faster. Building on this evidence, I develop a quantitative heterogeneous firm dynamics model in which financially constrained private firms choose whether and when to go public and how much equity to sell. An IPO provides an immediate capital injection and transitions the firm into a public entity with greater borrowing capacity and scalability. Because these benefits increase with firm-specific and aggregate productivity, the set of firms choosing to go public expands in booms. Calibrated to the U.S. nonfinancial firm sector, the model reproduces the cyclical behavior of IPO activity, capital raised, and post-IPO growth. The model shows that endogenous IPO activity amplifies and prolongs the response of aggregate output to contractionary shocks. Amplification arises because public firms adjust their scale more sharply, while persistence arises because downturns suppress IPO activity, creating missing cohorts that reduce the public firm share and alter its composition for many years.

JEL Classification: E22, G32, G34, D21

Keywords: IPO cyclicality, firm dynamics, financial frictions, business cycle

*Waseda Institute for Advanced Study, Waseda University; jisu.jeun@aoni.waseda.jp; Part of this paper was written at the Bank of Finland.

1 Introduction

The slow recovery following the Great Recession highlighted the macroeconomic importance of disruptions to firm financing. A large literature in macroeconomics studies financial frictions through borrowing constraints arising from collateral requirements or limited contract enforcement (Kiyotaki and Moore, 1997; Jermann and Quadrini, 2012; Khan and Thomas, 2013). Under such constraints, productive firms with insufficient collateral cannot finance their desired investment and therefore remain inefficiently small. Yet debt is not the only source of external finance. One such alternative is an initial public offering (IPO), through which a private firm gains access to public equity markets and can raise a large amount of capital in a single transaction. The IPO margin may be especially important for small, young, and productive firms, which contribute disproportionately to job creation and productivity growth but often face substantial financial constraints (Hadlock and Pierce, 2010; Haltiwanger et al., 2013; Decker et al., 2016). Abstracting from IPOs may therefore omit an important channel through which external finance shapes firm growth and aggregate fluctuations.

Going public is endogenous along several margins: firms choose whether to undertake an IPO, when to do so, and how much equity to sell. Unlike private equity financing or a seasoned equity offering, an IPO combines equity financing with a transition from private to public status. This distinction matters because private and public firms differ in their access to external finance, disclosure and compliance obligations, and ability to expand the scale of their operations (Brav, 2009; Borisov et al., 2021; Bernstein, 2022). Firm growth may therefore depend not only on the capital raised at listing but also on becoming public. Importantly for aggregate dynamics, IPO activity varies systematically over the business cycle.[1] This cyclicality can affect the aggregate economy through two channels. First, fewer firms go public in downturns, reducing the flow of public equity capital available to finance

[1]Existing work documents pronounced hot and cold IPO markets and links variation in IPO volume to aggregate capital demand and market conditions; see Lowry (2003); Pástor and Veronesi (2005); Ivanov and Lewis (2008); Tran and Jeon (2011).

investment and growth. Second, cyclical IPO activity changes the composition of private and public firms. Because the two types operate under different financing and production environments, this compositional shift may itself affect aggregate dynamics. Standard firm-dynamics models that abstract from endogenous transitions between private and public status leave this compositional channel largely unquantified.

This paper studies why IPO activity is cyclical and how this cyclicality shapes aggregate fluctuations. I first document that IPO activity covaries with the business cycle along three margins: the number of IPOs, the amount of capital raised at listing, and firms' post-IPO growth. The number of IPOs and capital raised are generally procyclical, whereas firms listing during contractions subsequently grow faster, suggesting cyclical selection into public status. To interpret these patterns and quantify their aggregate implications, I develop a quantitative heterogeneous firm dynamics model in which private firms choose whether and when to go public and how much equity to sell. An IPO provides an immediate capital injection and moves the firm into a public regime with greater borrowing capacity and scalability, which benefit the financially constrained firms the most. Because the value of these benefits increases with both firm-specific and aggregate productivity, the set of firms choosing to go public expands in booms, generating procyclical IPO activity and state-dependent selection. The calibrated model shows that the endogenous IPO margin amplifies and prolongs the response of aggregate output to shocks. Amplification reflects the greater responsiveness of public firms' desired scale, while persistence arises because downturns suppress IPO activity, creating missing cohorts that reduce the public firm share and alter its composition for many years.

Related literature. This paper contributes to three strands of the literature. First, this paper relates to a literature on cycles in IPO activity. IPO issuance is highly clustered over time, producing periods of unusually high and low issuance commonly referred to as hot

and cold IPO markets.[2] A large corporate-finance literature examines the forces underlying this clustering (Lowry, 2003; Pástor and Veronesi, 2005; Ivanov and Lewis, 2008). Lowry (2003), in particular, evaluates three potential explanations—private firms' demand for capital, investor sentiment, and adverse-selection costs—and finds capital demand and investor sentiment to be the most important, both statistically and economically. A related literature asks whether IPO cycles change the types of firms that go public and their subsequent performance. The evidence is mixed. Helwege and Liang (2004) find relatively modest differences in the observable characteristics of firms listing in hot and cold markets. Yung et al. (2008), however, document greater dispersion in long-run returns and higher delisting rates among hot-market IPOs, while Chemmanur and He (2011) find that firms listing during IPO waves have lower pre-IPO productivity and weaker post-IPO operating performance. IPO cycles may therefore affect not only the number of firms going public but also the composition of IPO cohorts. This paper relates IPO cycles explicitly to the aggregate business cycle. Closest to this perspective, Tran and Jeon (2011) study the dynamic relationships between macroeconomic conditions and the number of IPOs, total IPO proceeds, and average proceeds per IPO. Whereas their analysis characterizes reduced-form relationships between aggregate IPO measures and macro-financial conditions, I model firms' decisions over whether and when to go public and how much equity to sell. This approach decomposes fluctuations in capital raised into changes in valuations and in the fraction of equity sold, and it permits me to quantify how endogenous IPO decisions propagate aggregate shocks.

Second, this paper contributes to the literature on heterogeneity between private and public firms. Private firms constitute the vast majority of firms and account for a substantial share of aggregate activity, whereas public firms are fewer but disproportionately represented in the upper tails of the size and growth distributions. Both types are therefore important for aggregate firm dynamics. Davis et al. (2007) document pronounced differences in growth-rate

[2]The precise definition varies across studies. Early work identifies hot issue markets using unusually high initial returns (Ibbotson and Jaffe, 1975; Ritter, 1984), whereas later studies commonly classify markets using detrended IPO volume (Helwege and Liang, 2004; Alti, 2006).

volatility by listing status. One reason would be due to the selection into public status which is far from random. Larger and more productive firms with stronger growth opportunities are more likely to undertake an IPO (Pagano et al., 1998; Chemmanur et al., 2010). Evidence nevertheless indicates that selection does not fully account for the differences associated with listing status. Private firms rely more heavily on debt and maintain higher leverage than public firms (Brav, 2009), pay higher spreads on bank loans (Saunders and Steffen, 2011), and face lower borrowing rates after an IPO as public disclosure reduces lenders' informational advantages (Schenone, 2010), which is consequential to the subsequent growth, particularly in industries that depend more heavily on external finance (Borisov et al., 2021). Going public therefore changes not only the firm's ownership structure but also its financing and operating environment. Despite these differences, standard firm-dynamics models typically abstract from listing status as a dimension of firm heterogeneity.[3] A small recent literature jointly models private and public firms with an endogenous listing decision, but focuses primarily on the secular decline in U.S. public listings since around 2000 and its long-run determinants(González, 2020; Davydiuk et al., 2020; Casella et al., 2023) . I instead study how the private–public margin varies over the business cycle, how it changes the share and composition of public firms, and how these compositional changes propagate aggregate shocks.

Third, this paper contributes to the literature on financial frictions and capital allocation. A large body of work embeds financing constraints in models of firm dynamics, showing that they slow the growth of young and small firms and shape their investment, financing, and exit decisions (Cooley and Quadrini, 2001; Gomes, 2001; Albuquerque and Hopenhayn, 2004; Clementi and Hopenhayn, 2006). Because these constraints bind most tightly for productive firms with little net worth, they also distort the cross-sectional allocation of capital and depress aggregate productivity (Restuccia and Rogerson, 2008; Hsieh and Klenow, 2009; Buera et al., 2011; Midrigan and Xu, 2014; Gopinath et al., 2017). Moll (2014) shows that,

[3]Ottonello and Winberry (2020) themselves note that their Compustat sample excludes private firms, which are likely subject to more severe financial frictions.

although firms may eventually overcome borrowing constraints by accumulating internal funds, this process can be slow, making financial frictions particularly important during transitions and potentially over the business cycle.

The slow recovery following the Great Recession renewed interest in how such frictions shape aggregate fluctuations. Khan and Thomas (2013) show quantitatively that a tightening of borrowing conditions disrupts the reallocation of capital toward productive firms and can generate a recession that persists well beyond the underlying financial shock. Related work studies how financial shocks and fluctuations in uncertainty propagate through heterogeneous firms (Jermann and Quadrini, 2012; Buera and Moll, 2015; Arellano et al., 2019). Empirically, credit market disruptions had sizable effects on employment at financially constrained firms (Chodorow-Reich, 2014; Siemer, 2019), and the cyclical behavior of firms differs systematically with size (Gertler and Gilchrist, 1994; Crouzet and Mehrotra, 2020). These analyses underscore the importance of identifying the dimensions of firm heterogeneity that govern the transmission of aggregate shocks. I introduce listing status as one such dimension.

Most of this literature represents external finance through collateralized borrowing. Where equity issuance is modeled, it is typically treated as a continuous margin available to every firm at a proportional or fixed cost (Gomes, 2001; Hennessy and Whited, 2005, 2007; Di Nola, 2015), a structure that has been used to study the cyclical behavior of debt and equity finance (Covas and Den Haan, 2011; Begenau and Salomao, 2019). Going public differs in two respects: it is a discrete and largely irreversible decision, and it bundles a one-time capital injection with a permanent change in the firm's financing and operating environment. I model the IPO as such a transition: even when its borrowing constraint binds, a private firm can raise equity by going public, thereby financing investment that would otherwise be delayed and moving into a regime with looser borrowing limits. The model therefore allows me to study how endogenous access to public equity—and the cyclicality of that access—changes capital allocation and the propagation of aggregate shocks.

The remainder of the paper is organized as follows. Section 2 presents empirical evidence on the relationship between IPO activity and the business cycle. Section 3 develops the quantitative heterogeneous-firm model with endogenous IPO decisions. Section 4 uses the calibrated model to examine the mechanisms underlying these decisions and quantify how cyclical IPO activity propagates aggregate shocks.

2 Empirical Evidence

This section documents several dimensions along which IPO activity covaries with the business cycle. I begin by examining the aggregate extensive and intensive margins: the number of firms going public and the average proceeds raised per listing, respectively. I measure aggregate conditions using the cyclical component of the HP filtered quarterly real gross value added in the U.S. nonfarm business sector. The IPO sample is drawn from the merged CRSP–Compustat database. The resulting panel contains 71,529 firm-year observations for 7,541 firms that went public between 1980 and 2019.[4]

I first show that the number of IPOs is positively related to the business cycle, and that this relationship is concentrated in the period since 2000. I estimate

$$\log(1 + N_t^{IPO}) = \alpha + \beta g_t + \lambda Post_t + \theta(g_t \times Post_t) + \delta_1 t + \delta_2 t^2 + \varepsilon_t, \tag{1}$$

where N_t^{IPO} is the number of IPOs in quarter t, g_t is the output gap, the business-cycle measure defined above, and $Post_t$ equals one beginning in 2000Q1. The coefficient β captures the pre-2000 relationship, θ measures its change after 2000, and $\beta + \theta$ gives the post-2000 relationship. Given the sharp shift in IPO activity around 2000, the specification also allows

[4]Following standard sample restrictions, I retain U.S. nonfinancial and nonutility firms and exclude subsidiaries and firms involved in leveraged buyouts. I also exclude firms that delist because of a merger or acquisition within one year of becoming public. Because the analysis focuses on the subsequent growth of firms as independent public companies, this restriction reduces the influence of listings followed by an imminent change in control. It reduces the number of firms in the sample by 0.8 percent.

Table 1: Number of IPOs and the business cycle

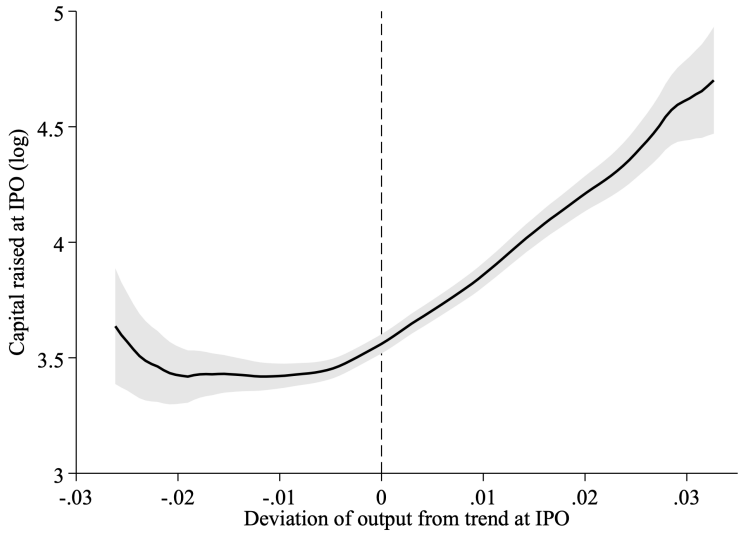
	(1)	(2)	(3)	(4)
Output gap (β)	0.230^{***}	0.084	0.053	0.027
	(0.085)	(0.081)	(0.070)	(0.069)
$\times$ Post-2000 (λ)			0.314^{***}	0.259^{**}
			(0.113)	(0.120)
Post-2000 (θ)			-3.122^{***}	-2.833^{***}
			(0.421)	(0.423)
log CAPE		2.590^{***}		0.954
		(0.598)		(0.598)
Quadratic time trend	✓	✓	✓	✓
Observations	160	160	160	160
R^2	0.366	0.501	0.717	0.731

the intercept to change and controls for a quadratic time trend.[5] Column (1) of Table 1 reports the parsimonious full-sample relationship. A 1 percentage point increase in year-over-year real gross value-added growth is associated with a 0.23 log point increase in IPO activity. Column (3) estimates equation (1). The pre-2000 relationship is close to zero, whereas the implied post-2000 coefficient is 0.31 and statistically significant. The estimates therefore indicate that IPO activity became substantially more sensitive to aggregate conditions after 2000.

Capital raised at the IPO also varies with the cycle. I measure capital raised by the sale of common and preferred stock in the IPO quarter, deflated to 2012 dollars and expressed in logs. Figure 1 plots a local linear estimate of the conditional mean of log capital raised at the IPO against the output gap at the time of the IPO. Capital raised increases with the output gap, but the relationship is nonmonotonic. As the output gap falls below zero, the decline flattens, and below approximately -0.02 the fitted relationship turns slightly upward. One possible explanation is that the composition of IPO firms changes over the

[5]Beginning around 2000, the average quarterly number of IPOs fell by more than half, and its time-series volatility also declined. The secular decline in IPO activity has been attributed in part to the higher fixed costs of public status following the enactment of the Sarbanes–Oxley Act in 2002. The literature also emphasizes changes in acquisition incentives, economies of scope, and the availability of private capital. See Gao et al. (2013); Bova et al. (2014); Doidge et al. (2017); Ewens and Farre-Mensa (2020); Ewens et al. (2024).

Figure 1: Capital Raised at IPO over the business cycle



cycle: in deep contractions, only firms with strong growth prospects may find it worthwhile to go public, which offsets the decline in average proceeds that weaker valuations would otherwise produce.

Finally, I show that firms that go public in different phases of the cycle follow systematically different growth paths after the IPO. I classify a firm to the contraction cohort if the output gap in its IPO quarter is negative, and to the expansion cohort otherwise. I then estimate the difference in post-IPO growth path between the two cohorts using the regression

$$\log k_{it} = \alpha + \sum_{a=1}^{A} \beta_a \mathbb{1}\{ipoage_{it} = a\} + \lambda Cohort_i + \sum_{a=1}^{A} \delta_a \mathbb{1}\{ipoage_{it} = a\} \times Cohort_i + \rho \log k_{i0} + \gamma y_t + \mu_{r(i)} + \nu_{n(i)} + \epsilon_{it}, \quad (2)$$

where $\log k_{it}$ is the post-IPO size of capital $ipoage_{it}$ denotes the number of years since firm i went public, and $cohort_i$ equals one for the contraction cohort and zero for the expansion cohort. The specification controls for the initial size at IPO k_{i0}, contemporaneous aggregate conditions y_t, headquarters-state fixed effects $\mu_{r(i)}$, and industry fixed effects $\nu_{n(i)}$. The coefficients δ_a trace the differential post-IPO capital path of the contraction cohort relative

to the expansion cohort, which is described in Table 2. In columns (1)–(3), the coefficients on the interaction between the contraction cohort indicator and IPO age are positive and statistically significant from the third year after the IPO, and remain large thereafter. That is, contraction cohort accumulates a larger capital stock relative to their size at the IPO.

Motivated by the nonmonotonic relationship between capital raised and the cycle, columns (4) and (5) split the sample by the size of the output gap at the IPO. Column (4) includes firms whose gap lies within ± 0.02, and column (5) includes those with a gap larger than 0.02 in absolute value, comparing firms that go public in deep contractions with those that go public in strong expansions. In column (4) the difference between cohorts emerges in years three to five and fades by year seven, whereas in column (5) it keeps growing with age, reaching 0.47 log points by year seven. Hence the early difference is concentrated among firms that go public near the middle of the cycle, while the persistent difference comes from the tails of the distribution. These patterns do not show that firms going public in deep contractions are inherently more productive. They are, however, consistent with the selection interpretation above: firms that go public during severe contractions may therefore be positively selected on growth prospects. The results are qualitatively similar when firm growth is measured using employment.

Overall, the evidence suggests that firms' IPO decisions vary systematically over the business cycle. More firms go public when aggregate conditions are favorable, and the amount of capital raised is generally procyclical, apart from the upturn observed in severe contractions. Firms that go public during contractions also tend to grow faster after listing. These patterns are consistent with positive selection into IPOs during downturns: only firms with sufficiently strong growth prospects may find it worthwhile to go public under unfavorable aggregate conditions. In the following section, I develop a heterogeneous-firm dynamics model with financial frictions and endogenous IPO decisions to investigate the mechanisms underlying these patterns and quantify their implications for aggregate fluctuations.

Table 2: Post-IPO Capital Accumulation by Listing Cohort

	(1)	(2)	(3)	(4)	(5)
Contraction	-0.022^{***}	0.015	-0.001	0.001	0.167^{**}
	(0.007)	(0.010)	(0.012)	(0.013)	(0.070)
Contraction × IPO age					
1 year	0.007	-0.017	-0.015	-0.013	-0.013
	(0.021)	(0.021)	(0.021)	(0.022)	(0.079)
3 years	0.199^{***}	0.138^{***}	0.141^{***}	0.105^{***}	0.179
	(0.036)	(0.038)	(0.037)	(0.040)	(0.125)
5 years	0.209^{***}	0.168^{***}	0.170^{***}	0.091^{*}	0.378^{**}
	(0.049)	(0.049)	(0.049)	(0.053)	(0.154)
7 years	0.158^{***}	0.131^{**}	0.129^{**}	0.027	0.473^{***}
	(0.058)	(0.058)	(0.058)	(0.063)	(0.162)
Initial capital	0.882^{***}	0.882^{***}	0.836^{***}	0.840^{***}	0.826^{***}
	(0.007)	(0.007)	(0.010)	(0.009)	(0.040)
Current aggregate state		2.093^{***}	2.041^{***}	2.166^{***}	1.779^{*}
		(0.375)	(0.370)	(0.397)	(1.033)
IPO age dummy	✓	✓	✓	✓	✓
State FE			✓	✓	✓
Industry FE			✓	✓	✓
Sample	All	All	All	$\lvert\text{gap}\rvert < 0.02$	$\lvert\text{gap}\rvert > 0.02$
Observations	43,617	43,617	43,617	38,330	5,287
R^2	0.748	0.748	0.758	0.758	0.802

3 Model

3.1 Environment

Time is discrete and the horizon is infinite. One unit of time is a quarter. Firms are either private or public, with type-specific characteristics described below. I first present the common structure of the firm problem and then introduce the distinctions across firm types.

A continuum of firms produces a homogeneous good using capital k_{it} and labor l_{it},

$$y_{it} = z_t s_{it} (k_{it}^{\alpha} l_{it}^{1-\alpha})^{\eta} \ , \tag{3}$$

where z_t denotes aggregate productivity and s_{it} firm-specific productivity. Decreasing returns to scale $\eta \in (0,1)$ ensures a finite optimal firm size. Aggregate productivity follows $\log z_{t+1} = \rho_z \log z_t + \varepsilon_{t+1}$ where $\varepsilon_{t+1} \sim N(0, \sigma_z^2)$, and idiosyncratic productivity follows $\log s_{i,t+1} = \rho_s \log s_{it} + \nu_{t+1}$ where $\nu_{t+1} \sim N(0, \sigma_s^2)$ and is independent of ε.

The timing within period t is as follows. First, the firm observes (z_t, s_{it}); second, it pays the fixed operating cost f, hire l_{it} with wage normalized at 1, and produces y_{it} with its predetermined capital k_{it}. As a result, operating profit is realized as $\pi_{it} = y_{it} - l_{it} - f$; third, both private and public firms may endogenously or exogenously exit. Private firms additionally decides whether to go public, as described in the following part. Finally, a firm that continues operating chooses next period's capital $k_{i,t+1}$.[6]

Capital depreciates at rate $\delta \in (0,1)$. A continuing firm finances the investment $k_{i,t+1} - (1-\delta)k_{it}$ out of operating profit net of debt service and one-period risk-free borrowing $b_{i,t+1}$. The loan is repaid at $t+1$ with interest at the risk-free rate r_t, defined below, and is subject to the collateral constraint

$$b_{i,t+1} \leq \theta \, k_{i,t+1}. \tag{4}$$

Since $\theta(1+r_t) \leq 1-\delta$, lenders are always repaid in full from the liquidation value of capital. Dividends are the residual cash flow,

$$d_{it} = \pi_{it} - (1+r_t)\, b_{it} + b_{i,t+1} - \left[k_{i,t+1} - (1-\delta)k_{it}\right] \ \geq \ 0. \tag{5}$$

[6]The model abstracts from endogenous wage dynamics to focus on the allocation of capital in the presence of IPOs and transition from private to public. Allowing wages to adjust in general equilibrium would introduce an additional propagation mechanism, which I leave for future work.

Dividends cannot be negative, which rules out costless equity issuance; a firm can issue equity only once, through an IPO, described in more detail below.

Firms discount future payoffs using the stochastic discount factor defined as

$$M(z_t, z_{t+1}) = \beta \exp\big(\phi_0 \log z_t + \phi_1 \log z_{t+1}\big), \tag{6}$$

following Clementi and Palazzo (2019). $\beta \in (0,1)$ is the time-preference parameter, $\phi_0 > 0$ shifts the level of the discount factor with current productivity, whereas $\phi_1 < 0$ determines how the discount factor varies across future productivity states, so that its magnitude governs the price of aggregate risk. Thus, the implied risk-free rate, $1 + r_t = 1/\mathbb{E}_t[M(z_t, z_{t+1})]$, is countercyclical if and only if $\phi_0 + \rho_z \phi_1 > 0$.

Exit and entry. Firms may exit, endogenously or exogenously. An exiting firm liquidates its cash on hand,

$$x_{it} \equiv \pi_{it} + (1-\delta)\, k_{it} - (1 + r_t)\, b_{it}. \tag{7}$$

A firm exits endogenously when x_{it} exceeds its expected continuation value, which is the entire stream of future dividends. In addition, a fraction ψ of firms exits exogenously each period, independently of type, and liquidates at x_{it} as well.

Each period, a constant mass of potential entrants, normalized to one, draws an initial idiosyncratic productivity $s_0 \sim G$. A potential entrant observes s_0 and enters if the expected discounted value of entering exceeds the entry cost c_e; upon entry, it installs capital k_0 and starts producing next period with productivity $s_1 = \rho_s s_0 + \sigma_s \varepsilon_1$. Because the value of entry is increasing in both s_0 and z_t, entry is characterized by a productivity cutoff $\bar{s}_0(z_t)$ that decreases with the aggregate state. Given the fixed mass of potential entrants, the mass of entering firms is therefore procyclical.

Firm types. A firm is private ($j = pr$) or public ($j = pb$). All firms are born as private, and a private firm may become public through an IPO. The transition from private to public is assumed to be irreversible.[7] The returns to scale, the collateral parameter, and the fixed operating cost depend on the firm's type,

$$\alpha = \alpha^j, \qquad \theta = \theta^j, \qquad f = f^j, \qquad j \in \{pr, pb\},$$

so that a type-j firm produces $y_{it}^j = z_t s_{it}(k_{it}^{\alpha} l_{it}^{1-\alpha})^{\eta^j}$ with $\eta^j \in (0,1)$, earns operating profit $\pi_{it^j} = y_{it}^j - l_{it} - f^j$, and faces the collateral constraint $b_{i,t+1} \leq \theta^j k_{i,t+1}$ with $\theta^j \in [0,1)$. An IPO therefore matters not only as a means of raising external finance but also as a transition from private to public status that changes the firm's financing and operating environment. Although the model imposes no ordering a priori, the calibration in Section 4 yields $\alpha^{pb} > \alpha^{pr}, \theta^{pb} > \theta^{pr}, f^{pb} > f^{pr}$. These estimates imply that public firms operate more efficiently at scale and have greater borrowing capacity, but incur higher fixed operating costs. In reduced form, these differences capture the improved access to debt markets associated with mandatory disclosure, as well as the costs of complying with disclosure and other listing requirements.

Another distinctive feature of public firms is that their dividends must be shared with outside shareholders. Investors who purchase equity at the IPO acquire a proportional claim to the firm's future distributions, including dividends and liquidation proceeds.[8] I describe how these claims enter the firm's problem.

Initial Public Offering. Going public takes two periods. If the IPO actually takes place at t, then a private firm initiates the IPO process at $t-1$ by paying a fixed cost κ^f and

[7] Delisting, i.e. going back to private is rare relative to IPOs except for certain period in early 2000s, and public firms leave the public sector mainly through acquisition or liquidation, which the model captures through exit.

[8] I do not model outside investors as solving a separate optimization problem. Both incumbent and outside shareholders value future cash flows using the common stochastic discount factor (M). The firm therefore maximizes the expected discounted stream of total dividends, while ownership shares determine how those dividends are divided among shareholders.

becomes a pending firm at t. A pending firm operates as a private firm and do not exit either endogenously or exogenously. At t, after observing (z_t, s_{it}), it decides whether to withdraw or to finalize the IPO. If it finalizes, the offer price reflects the fundamental value of the firm, i.e., the expected discounted value of the dividends to be generated as a public firm,

$$p(k_{it}, b_{it}, s_{it}, z_t) \tag{8}$$
$$= \mathbb{E}_t\Big[M(z_t, z_{t+1})\Big(\psi \max\{x^{pb}_{i,t+1}, 0\} + (1-\psi) \max\left\{d^{pb}_{i,t+1} + p(k_{i,t+1}, b_{i,t+1}, s_{i,t+1}, z_{t+1}),\ x^{pb}_{i,t+1}\right\}\Big)\Big].$$

Here, $d^{pb}_{i,t+1} + p(\cdot)$ is set to zero whenever $x^{pb}_{i,t+1} < 0$, since such a firm cannot continue. Shares are sold at $\hat{p}_{it} = q_t\, p(k_{it}, b_{it}, s_{it}, z_t)$, where $q_t \in (0, 1]$ is a common, economy-wide factor that discounts the fundamental value of the firm and whose determination is described below. Given $\hat{p}_{it}$, the firm chooses the share $\chi \in (0, 1)$ to sell and raises $\chi\, \hat{p}_{it}$. Upon finalization, the firm pays the underwriters a fee equal to a fraction κ of the proceeds, so that the capital injection via IPO is $(1-\kappa)\, \chi\, \hat{p}_{it}$. Alternatively, a firm that withdraws reverts to private status, neither raising capital by selling equity nor paying the underwriters' fee, and may reinitiate in a later period by paying κ^f again.

Each firms takes q_t as given, which is determined in equilibrium by the aggregate volume issued. Let $\mathcal{F}_t$ be the set of pending firms that finalize at t, let Γ^{ipo}_t denote the distribution of IPO firms, and define the aggregate fundamental value of newly issued equity,

$$Q_t \ \equiv \ \int_{i \in \mathcal{F}_t} \chi_i\, p(k_{it}, b_{it}, s_{it}, z_t)\, d\Gamma^{ipo}_t. \tag{9}$$

Investors absorb this volume at discount $1 - q_t$ according to

$$Q_t = \mathcal{S}(1 - q_t), \tag{10}$$

where $\mathcal{S}$ is a reduced-form demand curve for newly issued shares, which is continuous and

strictly increasing, with $\mathcal{S}(0) = 0$.[9] The frictionless benchmark $q_t \equiv 1$ corresponds to $\mathcal{S}$ infinitely elastic at zero discount. In particular, I specify the functional form as

$$\mathcal{S}(1 - q_t) = \bar{S}\,(1 - q_t)^{\gamma}, \qquad \bar{S} > 0,\ \gamma > 0, \tag{11}$$

so that the equilibrium discount is $q_t = 1 - (Q_t/\bar{S})^{1/\gamma}$. The scale $\bar{S}$ governs the size of the market and γ the elasticity of absorbed volume with respect to the discount. The model requires $\bar{S} > Q_t$ so that $q_t > 0$.

3.2 Value functions

For recursive formulation, time subscripts and the firm index i are dropped in this section, and next-period variables are denoted by a prime; the type superscript $j \in \{pr, pb\}$ is retained. A firm's state is in principle (k, b, s, z) together with the interest rate r_{-1} at which its debt was contracted. Capital, debt, and r_{-1} matter for the current decision only through cash on hand, so the state reduces to (x, s, z).

Public firms. A public firm with cash on hand $x \geq 0$ exits exogenously with probability ψ, in which case it is liquidated and the owners receive their share $(1-\chi)x$. With probability $1 - \psi$, the firm either continues or exits endogenously, whichever yields the higher value to the owners. The value of a public firm is therefore

$$V^{pb}(x, s, z; \chi) = \psi\,(1 - \chi)\,x + (1 - \psi)\,\max\{\tilde{V}^{pb}(x, s, z; \chi), (1 - \chi)x\}, \tag{12}$$

and $V^{pb}(x, s, z; \chi) = 0$ for $x < 0$, since a firm with negative cash on hand is forced to exit under limited liability. Here, a continuing public firm chooses next period's capital k' and borrowing b' to maximize the owners' share of the dividend plus the expected discounted

[9] This is motivated by evidence that demand curves for stocks slope downward, i.e., investor demand is considerably less elastic than frictionless asset-pricing models imply. See the evidence in U.S. equity markets documented in Gabaix and Koijen (2021) and Koijen and Yogo (2019).

continuation value,

$$\tilde{V}^{pb}(x,s,z;\chi) = \max_{k',b'} (1-\chi)\, d + \mathbb{E}\big[M(z,z')\, V^{pb}(x',s',z';\chi) \mid s,z\big], \tag{13}$$

subject to

$$d = x - k' + b' \geq 0, \qquad b' \leq \theta^{pb} k', \qquad x' = z' s' k'^{\alpha^{pb}} - f^{pb} + (1-\delta)k' - (1+r(z))\, b'.$$

Private firms. A private firm with cash on hand $x \geq 0$ exits exogenously with probability ψ and is liquidated at x. Otherwise, it exits endogenously or continues, whichever yields the higher value. Its value is

$$V^{pr}(x,s,z) = \psi\, x + (1-\psi) \max\{\tilde{V}^{pr}(x,s,z),\, x\}, \tag{14}$$

and $V^{pr}(x,s,z) = 0$ for $x < 0$. A continuing private firm either remains private or starts an IPO,

$$\tilde{V}^{pr}(x,s,z) = \max\{\tilde{V}^{stay}(x,s,z),\, \tilde{V}^{pend}(x,s,z)\}. \tag{15}$$

A firm that remains private chooses (k', b') and carries on as a private firm,

$$\tilde{V}^{stay}(x,s,z) = \max_{k',b'}\ d + \mathbb{E}\big[M(z,z')\, V^{pr}(x',s',z') \mid s,z\big], \tag{16}$$

subject to

$$d = x - k' + b' \geq 0, b' \leq \theta^{pr} k', x' = z' s' k'^{\alpha^{pr}} - f^{pr} + (1-\delta)k' - (1+r(z))\, b'.$$

A firm that starts an IPO pays the fixed cost κ^f and becomes a pending firm in the next

period,

$$\tilde{V}^{pend}(x,s,z) = \max_{k',b'} \; d + \mathbb{E}\big[M(z,z')\,V^{ipo}(x',s',z') \mid s,z\big], \tag{17}$$

subject to the same constraints, except that $d = x - \kappa^f - k' + b' \geq 0$.

A pending firm operates as a private firm and cannot exit. It either finalizes the IPO or withdraws,

$$V^{ipo}(x,s,z) = \max\{\tilde{V}^{fin}(x,s,z),\, \tilde{V}^{wd}(x,s,z)\}, \quad x \geq 0, \tag{18}$$

and $V^{ipo}(x,s,z) = 0$ for $x < 0$. A firm that finalizes chooses the share χ of equity to sell, together with (k',b'),

$$\tilde{V}^{fin}(x,s,z) = \max_{k',b',\chi} \; d + \mathbb{E}\big[M(z,z')\,V^{pb}(x',s',z';\chi) \mid s,z\big], \tag{19}$$

subject to

$$d = x + (1-\kappa)\,\chi\,q_t\,p(x,s,z) - k' + b' \geq 0, \qquad b' \leq \theta^{pb}k', \qquad x' = z's'k'^{\alpha^{pb}},$$

where the price can be expressed using value function of public firms as

$$p(x,s,z) = \mathbb{E}_t\Big[V^{pb}(x',s',z';\chi) \mid s,z\Big]. \tag{20}$$

Potential entrants. A potential entrant observes z and draws its initial idiosyncratic productivity s from G, the stationary distribution of the idiosyncratic process. It is endowed with net worth x_0 and cannot borrow at the entry stage. After paying the entry cost c_e, it chooses next period's capital k' to maximize the expected discounted value of operating as

a private firm,

$$V^e(s,z) = \max_{k'} \; d + \mathbb{E}\big[M(z,z')\,V^{pr}(x',s',z') \mid s,z\big], \tag{21}$$

subject to $d = x_0 - k' \geq 0$ and $x' = \pi'^{pr} - f^{pr} + (1-\delta)k'$. The entrant enters if $V^e(s,z) \geq c_e$, so that the mass of entrants is $N_t = \int \mathbb{1}\{V^e(s,z_t) \geq c_e\}\, dG(s)$.

3.3 Competitive equilibrium

Let $\Gamma_t = (\Gamma_t^{pr}, \Gamma_t^{ipo}, \Gamma_t^{pb})$ denote the distribution of firms over private, pending, and public status, where Γ_t^{pr} and Γ_t^{ipo} are measures over (x,s) and Γ_t^{pb} is a measure over (x,s,χ). The aggregate state is (z,Γ). Since $r(z)$ and $M(z,z')$ are determined by z alone, the distribution matters for individual decisions only through the offer discount q_t.

A *recursive competitive equilibrium* consists of value functions $\{V^{pr}, V^{ipo}, V^{pb}\}$ and continuation values $\{\tilde{V}^{pr}, \tilde{V}^{stay}, \tilde{V}^{pend}, \tilde{V}^{fin}, \tilde{V}^{wd}, \tilde{V}^{pb}\}$; policy functions for capital and borrowing (k', b') of each firm type, the offer share χ, and the decisions to start an IPO, finalize or withdraw, and exit; an offer price $p(x,s,z)$; a discount function $q(z,\Gamma)$, with $q_t = q(z_t,\Gamma_t)$; a mass of entrants N_t; distributions $\{\Gamma_t^{pr}, \Gamma_t^{ipo}, \Gamma_t^{pb}\}$ and a law of motion $\Gamma_{t+1} = H(z_t, \Gamma_t, z_{t+1})$, such that

(i) *Firm optimization.* Given $q(z,\Gamma)$ and H, the value and policy functions solve the Bellman equations of private, pending, and public firms, subject to the feasibility and collateral constraints.

(ii) *Offer price.* $p(x,s,z)$ satisfies (11), given the public firm's value and policies.

(iii) *Entry.* A potential entrant with productivity s enters if and only if $V^e(s,z) \geq c_e$, and the mass of entrants is $N_t = \int \mathbb{1}\{V^e(s,z_t) \geq c_e\}\, dG(s)$.

(iv) *Market for newly issued equity.* In every period, q_t solves (14): $\mathcal{Q}(q_t; z_t, \Gamma_t) = \mathcal{S}(1-q_t)$, where $\mathcal{Q}$ is the value of equity offered by firms that finalize.

(v) *Consistency.* H is generated by the policy functions, the exogenous processes for (z, s), and the entry mass N_t. Firms that start an IPO in $t-1$ form Γ_t^{ipo} (after forced exit of those with $x < 0$). In t, pending firms either finalize and join Γ_{t+1}^{pb} with the offer share χ they chose, or withdraw and rejoin Γ_{t+1}^{pr}. Public firms never return to Γ^{pr} or Γ^{ipo}. Entrants that enter in t join Γ_{t+1}^{pr} with the state (x', s') implied by their choice of k'.

Relative to the baseline equilibrium without a friction in the market for newly issued equity, only condition (iv) is new. The interest rate $r(z)$, the stochastic discount factor $M(z, z')$, and the collateral constraints (3) are unchanged. A recursive competitive equilibrium consists of value functions $\{V^{pr}, V^{pend}, V^{pb}\}$, policy functions for a', χ, and the initiation, finalization, exit, and entry decisions, an offer price $p(\cdot)$ satisfying (11), a sequence $\{q_t\}$, and measures $\{\Gamma_t^{pr}, \Gamma_t^{ipo}, \Gamma_t^{pb}\}$, such that: (i) firms optimize given $\{q_t\}$; (ii) entrants optimize and free entry holds; (iii) q_t solves (14) in every period; and (iv) the measures evolve according to the policy functions and the exogenous processes, with pending firms from $t-1$ either finalizing (joining Γ^{pb}) or withdrawing (returning to Γ^{pr}).

4 Quantitative Analysis

4.1 Calibration

I calibrate the model using data on U.S. nonfinancial firms, unless otherwise noted, from 2000Q1 through 2019Q4. A model period is one quarter, and each production unit corresponds to a firm. Table 3 reports the parameters assigned and calibrated externally. I set the depreciation rate, the capital share, and the private-firm returns-to-scale parameter to standard values from the macroeconomics literature. The proportional cost of issuing equity at an IPO κ is set to 7 percent, matching the average underwriting spread in the FactSet data. The specification of stochastic discount factor in equation (6) implies constant maximal Sharpe ratio, $\sqrt{\exp(\gamma_1^2\sigma_z^2) - 1}$. I choose $(\beta, \gamma_0, \gamma_1)$ to match three quarterly moments: the mean (0.4 percent) and standard deviation (0.9 percent) of the real risk-free rate and

the Sharpe ratio of the aggregate stock market (0.22), following Clementi and Palazzo (2019).

Table 3: Assigned parameters

Parameter	Description	Value
δ	Depreciation rate	0.025
η	Returns to scale	0.80
α	Capital share	0.30
κ	Underwriting fee	0.07
β	Time preference	0.973
ϕ_0	Pricing kernel	9.26
ϕ_1	Pricing kernel	-12.08

The remaining parameters are calibrated jointly to the moments in Table 4. Although all parameters affect all moments, each is tied most closely to one, so the stationary moments match their empirical counterparts. The estimator is simulated method of moments with weighting of inverse of the moments' variance. The productivity processes are calibrated in two steps. First, I shut down aggregate fluctuations by setting $z = 1$ and calibrate the idiosyncratic process, (ρ_s, σ_s), in the stationary equilibrium to moments of the firm-level investment rate. These moments are those reported by Cooper and Haltiwanger (2006) for a balanced panel of manufacturing plants in the Longitudinal Research Database over 1972–1988, which Clementi and Palazzo (2016) also target. Second, holding the idiosyncratic process fixed, I calibrate the aggregate productivity process (ρ_z, σ_z) to the moments of output growth which I use nonfarm private value added data from the Bureau of Economic Analysis. The collateral parameters θ^{pb} and θ^{pr} match debt-to-asset ratios of public and all firms, respectively. For public firms, I use Compustat and compute the ratio as total debt (long-term plus current) divided by total assets. For private firm moment, I use the Financial Accounts of the United States (Flow of Funds) for nonfinancial corporate business. The operating costs f^{pb} and f^{pr} match quarterly exit rates. For public firms, the exit rate is 0.7 percent, computed as the number of delistings other than mergers divided by the number of listed firms; for all firms, it is 2.0 percent, from the Business Dynamics Statistics.

The parameters described so far jointly shape firms' decisions to go public. Given these parameters, the fixed cost of going public, κ^f, pins down the share of public firms. Although public firms account for only a small fraction of all firms, they account for a large share of employment, about 27 percent. To reconcile these two facts, returns to scale are allowed to differ between private and public firms, $\eta^{pb} = \eta^{pr} + \Delta\eta$. A higher $\Delta\eta$ raises the optimal scale of public firms relative to private firms and hence their share of employment, without changing the number of firms that go public. The premium $\Delta\eta$ is therefore chosen to match the employment share of public firms. Finally, the equity market demand is disciplined by two moments from FactSet IPO data: the average IPO discount, [] percent, and the elasticity of the discount to IPO volume, 0.69, estimated from a log-log regression on quarterly data. The equity-market friction is governed by the level $\bar{S}$ and the elasticity η_S of investor demand for newly issued shares, $Q_t = \bar{S}(1 - q_t)^{\eta_S}$. In the model, the first-day return of an IPO is $1/q_t - 1$: shares are sold at $q_t p$ and trade at their fundamental value p thereafter. The level $\bar{S}$ determines the discount at the average issuance volume and is chosen to match the mean first-day return of U.S. IPOs over 2001–2019, 14.7 percent (data from Ritter). The elasticity γ determines how much the discount moves with issuance volume over the cycle and is chosen to match the standard deviation of the annual average first-day return, 4.5 percentage points. Because the discount depends on aggregate issuance, firms must forecast it when deciding whether to go public; they use the rule $q_t = q(z_t)$, which is solved jointly with the remaining parameters as a fixed point of the simulated economy. The calibrated elasticity implies that a 1 percent increase in issuance volume raises the discount by 0.75 percent. In equilibrium, the discount ranges from 2.4 percent in the lowest productivity state to 16.6 percent in the highest, as IPOs cluster in booms.

4.2 IPO decision

This section assesses how well the calibrated model reproduces the empirical patterns in IPO activity introduced in section 2 and examines the underlying mechanisms. Figure 2

Table 4: Internally calibrated parameters

Parameter	Description	Value	Target moment	Data	Model
	Idiosyncratic productivity		Investment rate		
ρ_s	persistence	0.95	mean	0.122	0.118
σ_s	volatility	0.04	standard deviation	0.337	0.357
	Aggregate productivity		Output growth		
ρ_z	persistence	0.82	autocorrelation	0.063	0.064
σ_z	volatility	0.008	standard deviation	0.032	0.036
	Collateral constraint		Debt-to-asset ratio		
θ^{pb}	public firms	0.34	public firms	0.300	0.313
θ^{pr}	private firms	0.15	aggregate	0.223	0.169
	Operating cost		Exit rate		
f^{pb}	public firms	0.22	public firms	0.007	0.010
f^{pr}	private firms	0.11	aggregate	0.020	0.016
Δ_α	Returns to scale premium	0.015	Emp. share of public	0.273	0.166
κ^f	IPO fixed cost	0.3	Share of public firms	0.008	0.025
	Equity market demand		IPO discount		
$\bar{\mathcal{S}}$	Level	54.02	Mean first-day return	0.147	0.143
γ	Elasticity	1.33	S.D. of annual mean first-day return	0.045	0.046

compares the demeaned number of IPOs in the model and the data from 2000 onward. To construct the model series, I feed the observed HP-filtered cyclical component of real gross value added into the aggregate state z_t, and simulate IPO activity along the resulting path from its stationary distribution.

The model reproduces procyclical IPO activity through a state-dependent IPO policy. Figure 3 displays the IPO policy as a function of firm-specific productivity and size measured as net worth, capital net debt, at low, median, and high levels of aggregate productivity. The marked regions identify firms that choose to go public, and its color indicates the fraction of equity sold at the IPO, χ. As aggregate productivity rises, the IPO region expands, increasing the mass of firms that go public and generating the procyclical IPO activity documented above. The model also characterizes selection into public status. Across aggregate states,

Figure 2: Simulated number of IPOs

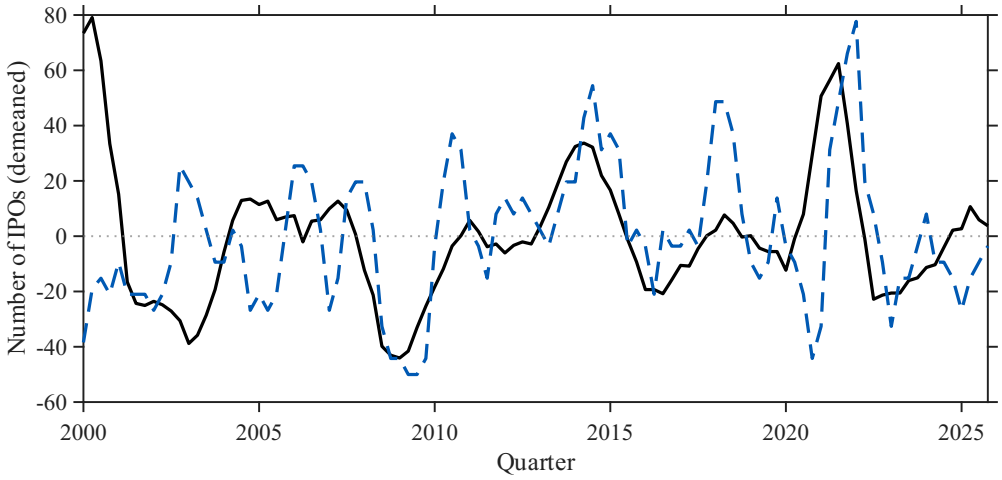
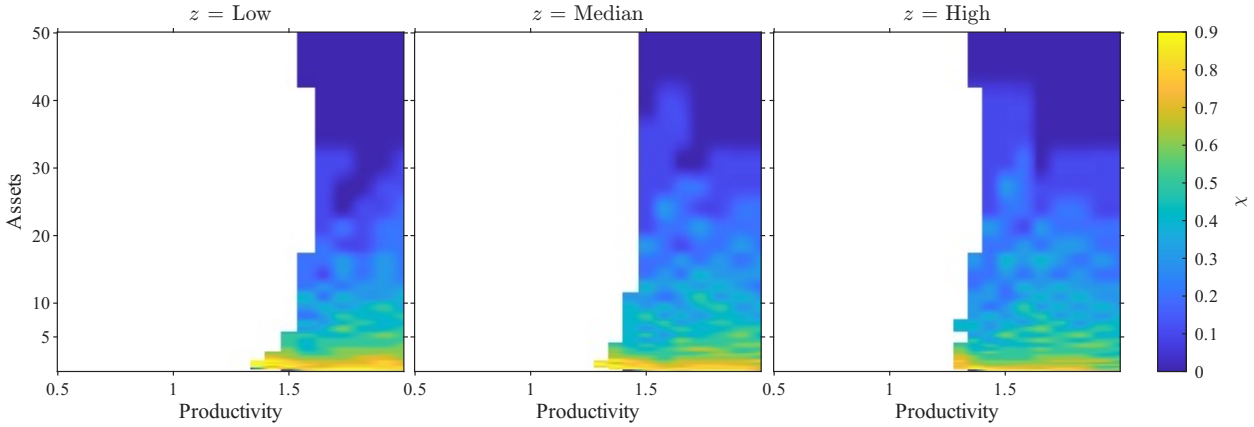


Figure 3: IPO policy function



firms that go public tend to be most productive but relatively small. These firms derive particularly large benefits from an IPO because financing constraints prevent them from expanding toward their efficient scale. Moreover, among firms that go public, smaller firms generally choose a larger χ. The choice of χ reflects an intertemporal substitutional trade-off between current financing and future dilution. Selling a larger equity share provides a larger immediate capital injection, allowing these firms to expand rapidly and exploit their high productivity, which outweighs the cost of distributing a larger share of future distributions to outside shareholders.

Furthermore, the model implies that this IPO selection varies over the business cycle. When aggregate productivity is low, less productive firms choose to delay their IPOs. There-

fore, firms that nevertheless go public in such states are more positively selected on productivity, making the average productivity of IPO firms countercyclical. This selection has implications for firms' subsequent growth. As documented in Section 2, firms that go public during contractions grow faster post-IPO. The model reproduces this pattern. The unconditional results in Table 5 compare the average post-IPO capital growth of firms listing at different levels of z. Firms that go public at low z subsequently grow faster than those that go public at high z, reflecting stronger positive selection on idiosyncratic productivity in low-z states. The model implies that, holding firm characteristics fixed, a firm that goes public at high z receives a higher offer price. Because the capital injection depends on both the offer price and the fraction of equity sold, this higher price generates greater proceeds unless it is offset by a sufficiently large reduction in the equity share sold. The resulting difference in capital injections can in turn affect post-IPO growth. With the endogenous adjustment of the equity-issuance discount q_t, however, this financing effect is partly attenuated.

To isolate the financing effect of IPO timing from selection, the conditional results restrict the sample to firm types that would choose to go public regardless of the aggregate state. Holding the composition of IPO firms fixed reverses the sign of the growth differential: firms that go public at high z subsequently grow faster than those that go public at low z. Unlike the unconditional differential, however, this high-z growth advantage diminishes with IPO age, suggesting that the effect of the initial capital injection is transitory and gradually dissipates. This pattern is qualitatively consistent with Borisov et al. (2021), who find that the effect IPO proceeds is short-lived. Thus, the financing effect favors firms listing at high z in the short run, whereas cyclical selection accounts for the faster and more persistent growth of low-z cohorts in the unconditional comparison.

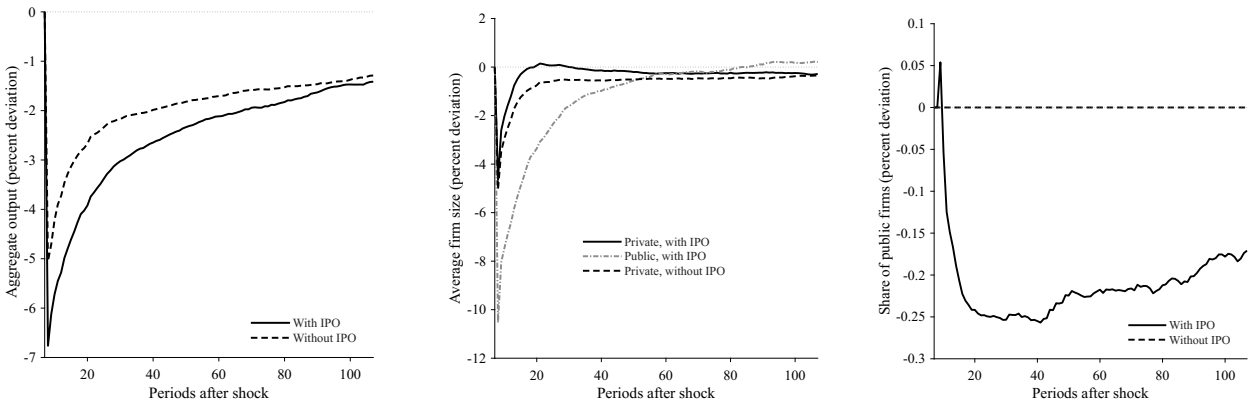
Statistic	1 year post IPO	3 years	5 years	7 years
Unconditional	0.77	4.43	10.90	12.53
Conditional	-18.08	-13.16	-7.60	-6.95

Table 5: Post-IPO capital growth differential: low-z minus high-z cohorts (percent)

4.3 Role of cyclical IPOs

In this section, I quantify the contribution of cyclical IPO activity to aggregate fluctuations. To this end, I compute simulation-based generalized impulse responses to an aggregate productivity shock. In each replication, I initialize the economy at its stationary distribution and simulate a 40-quarter pre-shock history of aggregate productivity drawn from the Markov chain for z. The shocked and unshocked paths share this history. At $t = 0$, I impose a one-time, unanticipated negative innovation to aggregate productivity that lowers TFP by 1 percent on impact. Thereafter, the two paths share the same aggregate and idiosyncratic innovations, so their difference isolates the effect of the aggregate shock from simulation noise. I simulate the baseline economy with endogenously cyclical IPOs and a no-IPO counterfactual in which κ is set high enough to prevent private firms from doing IPO; all other parameters remain unchanged. As no new firms transition from private to public status, al firms stay private in the counterfactual model. The difference between the impulse responses in the two economies measures the contribution of the endogenous IPO margin to the amplification and propagation of the contractionary shock.

Figure 4 compares the impulse responses to a negative productivity shock in the baseline economy with IPOs and in the counterfactual economy without IPOs. Panel (a) shows that aggregate output falls in both economies, but the decline is larger and more persistent in the baseline economy. First, the amplified initial decline reflects the presence of public firms. Firms that go public are productive firms whose growth was not realized due to the collateral constraint; the equity raised at the IPO and the looser borrowing limit of public firms allow them to operate close to their unconstrained scale after the IPO. Their capital therefore adjusts to the unconstrained optimum, which falls immediately with aggregate productivity.



(a) Aggregate Output e (b) Average Firm Size (c) Share of Public Firms

Figure 4: Impulse responses to a negative productivity shock

In addition, the higher returns to scale of public firms make their optimal scale more elastic to productivity. Hence, the average size of public firms falls by about twice as much as that of private firms.

Next, two forces contribute to the greater persistence of the shock in the baseline economy. First, the shock reduces the stock of public firms. Panel (c) shows that the public firm share rises slightly on impact, as firms already in the IPO pipeline complete their offerings, but subsequently falls by approximately 0.25 percentage points. This decline reflects a drop of about 10 percent in IPO initiations following the shock. Although the IPO rate eventually returns to its steady-state level, it does not overshoot sufficiently to replace the missing IPO cohorts. The public firm share therefore remains below its steady-state level throughout the horizon shown. Second, the average size of public firms recovers more slowly than that of private firms. This persistence partly reflects changes in the cohort composition of the public sector. Newly public firms contribute disproportionately to public-sector growth: they are positively selected on productivity and receive an equity injection that allows them to finance investment and expand toward their efficient scale. A contraction reduces the inflow of these firms and, conditional on firm characteristics, lowers the capital raised by those that still go public. The public firm consequently contains a persistently smaller share of young, fast-growing firms, slowing the recovery of average public firm size and adjusts only

gradually. This mechanism is analogous to the cohort effects emphasized by Sedláček and Sterk (2017), who show that aggregate conditions at entry alter the composition and growth potential of entering firms, leaving persistent effects as the affected cohorts age. Together, the lower public firm share and the slow recovery in public firm size prolong the response of aggregate output.

	std of $\log(MPK)$	
Model	Private	Public
With IPO	0.33	0.26
Without IPO	0.35	-

Table 6: Capital misallocation in steady state

Taken together, the impulse responses show that the IPO margin amplifies aggregate shocks and makes their effects more persistent. Amplification arises because public firms, whose desired scale is more sensitive to aggregate productivity, adjust their capital more sharply following a shock. Persistence arises because IPO activity is procyclical: a downturn reduces the flow of firms into public status, leaving a smaller public sector whose size and composition recover only gradually. These dynamic effects coexist with a steady-state allocative gain. Table 6 reports capital misallocation, measured by the variance of the log marginal product of capital, in the stationary equilibria of the two economies. Misallocation among private firms is lower in the baseline economy than in the economy without IPOs because productive firms whose expansion is most constrained by their net worth select into public status. Misallocation is lower still among public firms, as the equity raised at the IPO and the looser borrowing limit allow them to operate closer to their unconstrained scale. The option to go public therefore improves the steady-state allocation of capital and raises aggregate output, while amplifying and prolonging aggregate fluctuations.

5 Conclusion

This paper studies why IPO activity varies over the business cycle and how this cyclicality shapes aggregate fluctuations. I document that IPO activity covaries with aggregate conditions along three margins. The number of IPOs and the amount of capital raised at listing are generally procyclical, while firms that go public during contractions subsequently grow faster. Together, these patterns suggest that aggregate conditions affect not only the frequency and financing of IPOs but also the composition of firms that choose to go public.

To interpret these facts, I develop a quantitative heterogeneous-firm model in which financially constrained private firms choose whether and when to go public and how much equity to sell. An IPO provides an immediate capital injection and moves the firm into a public regime with greater borrowing capacity and scalability. In expansions, higher valuations and stronger investment incentives broaden the set of firms for which an IPO is profitable. In contractions, fewer firms go public, but those that do are positively selected on productivity and growth potential. The calibrated model reproduces the cyclical behavior of IPO activity and the faster post-IPO growth of contraction cohorts.

The model implies that the IPO margin has distinct implications for the level and dynamics of aggregate activity. The option to go public improves the allocation of capital by allowing productive, financially constrained firms to expand toward their efficient scale. At the same time, endogenous IPO activity amplifies and prolongs the effects of contractionary shocks. Public firms adjust their scale more sharply, and the decline in IPO activity creates missing cohorts that reduce the public firm share and alter its composition for many years. More broadly, these findings show that fluctuations in firms' access to public equity markets affect aggregate dynamics not only through the flow of external finance but also through persistent changes in the composition of private and public firms.

References

Albuquerque, Rui and Hugo A. Hopenhayn, "Optimal Lending Contracts and Firm Dynamics," *Review of Economic Studies*, 2004, *71* (2), 285–315.

Alti, Aydoğan, "How Persistent Is the Impact of Market Timing on Capital Structure?," *Journal of Finance*, 2006, *61* (4), 1681–1710.

Arellano, Cristina, Yan Bai, and Patrick J. Kehoe, "Financial Frictions and Fluctuations in Volatility," *Journal of Political Economy*, 2019, *127* (5), 2049–2103.

Begenau, Juliane and Juliana Salomao, "Firm Financing over the Business Cycle," *Review of Financial Studies*, 2019, *32* (4), 1235–1274.

Bernstein, Shai, "The Effects of Public and Private Equity Markets on Firm Behavior," *Annual Review of Financial Economics*, 2022, *14*, 295–318.

Borisov, Alexander, Andrew Ellul, and Merih Sevilir, "Access to Public Capital Markets and Employment Growth," *Journal of Financial Economics*, 2021, *141* (3), 896–918.

Bova, Francesco, Miguel Minutti-Meza, Gordon Richardson, and Dushyantkumar Vyas, "The Sarbanes-Oxley Act and Exit Strategies of Private Firms," *Contemporary Accounting Research*, 2014, *31* (3), 818–850.

Brav, Omer, "Access to Capital, Capital Structure, and the Funding of the Firm," *Journal of Finance*, 2009, *64* (1), 263–308.

Buera, Francisco J. and Benjamin Moll, "Aggregate Implications of a Credit Crunch: The Importance of Heterogeneity," *American Economic Journal: Macroeconomics*, 2015, *7* (3), 1–42.

_ , Joseph P. Kaboski, and Yongseok Shin, "Finance and Development: A Tale of Two Sectors," *American Economic Review*, 2011, *101* (5), 1964–2002.

Casella, Sara, Hanbaek Lee, and Sergio Villalvazo, "Disclosure Regulation, Intangible Capital and the Disappearance of Public Firms," Finance and Economics Discussion Series 2023-050, Board of Governors of the Federal Reserve System 2023.

Chemmanur, Thomas J. and Jie He, "IPO Waves, Product Market Competition, and the Going Public Decision: Theory and Evidence," *Journal of Financial Economics*, 2011, *101* (2), 382–412.

_ , Shan He, and Debarshi K. Nandy, "The Going-Public Decision and the Product Market," *Review of Financial Studies*, 2010, *23* (5), 1855–1908.

Chodorow-Reich, Gabriel, "The Employment Effects of Credit Market Disruptions: Firm-Level Evidence from the 2008–9 Financial Crisis," *Quarterly Journal of Economics*, 2014, *129* (1), 1–59.

Clementi, Gian Luca and Berardino Palazzo, "Entry, Exit, Firm Dynamics, and Aggregate Fluctuations," *American Economic Journal: Macroeconomics*, 2016, *8* (3), 1–41.

_ and _ , "Investment and the Cross-Section of Equity Returns," *Journal of Finance*, 2019, *74* (1), 281–321.

_ and Hugo A. Hopenhayn, "A Theory of Financing Constraints and Firm Dynamics," *Quarterly Journal of Economics*, 2006, *121* (1), 229–265.

Cooley, Thomas F. and Vincenzo Quadrini, "Financial Markets and Firm Dynamics," *American Economic Review*, 2001, *91* (5), 1286–1310.

Cooper, Russell W. and John C. Haltiwanger, "On the Nature of Capital Adjustment Costs," *Review of Economic Studies*, 2006, *73* (3), 611–633.

Covas, Francisco and Wouter J. Den Haan, "The Cyclical Behavior of Debt and Equity Finance," *American Economic Review*, 2011, *101* (2), 877–899.

Crouzet, Nicolas and Neil R. Mehrotra, "Small and Large Firms over the Business Cycle," *American Economic Review*, 2020, *110* (11), 3549–3601.

Davis, Steven J., John Haltiwanger, Ron Jarmin, and Javier Miranda, "Volatility and Dispersion in Business Growth Rates: Publicly Traded versus Privately Held Firms," *NBER Macroeconomics Annual*, 2007, *21*, 107–180.

Davydiuk, Tetiana, Brent Glover, and Rachel Szymanski, "The Decline in Public Firms," 2020. Working paper, Carnegie Mellon University.

Decker, Ryan A., John Haltiwanger, Ron S. Jarmin, and Javier Miranda, "Where Has All the Skewness Gone? The Decline in High-Growth (Young) Firms in the U.S.," *European Economic Review*, 2016, *86*, 4–23.

Doidge, Craig, G. Andrew Karolyi, and René M. Stulz, "The U.S. Listing Gap," *Journal of Financial Economics*, 2017, *123* (3), 464–487.

Ewens, Michael and Joan Farre-Mensa, "The Deregulation of the Private Equity Markets and the Decline in IPOs," *Review of Financial Studies*, 2020, *33* (12), 5463–5509.

_ , Kairong Xiao, and Ting Xu, "Regulatory Costs of Being Public: Evidence from Bunching Estimation," *Journal of Financial Economics*, 2024, *153*, 103775.

Gabaix, Xavier and Ralph S. J. Koijen, "In Search of the Origins of Financial Fluctuations: The Inelastic Markets Hypothesis," NBER Working Paper 28967, National Bureau of Economic Research 2021.

Gao, Xiaohui, Jay R. Ritter, and Zhongyan Zhu, "Where Have All the IPOs Gone?," *Journal of Financial and Quantitative Analysis*, 2013, *48* (6), 1663–1692.

Gertler, Mark and Simon Gilchrist, "Monetary Policy, Business Cycles, and the Behavior of Small Manufacturing Firms," *Quarterly Journal of Economics*, 1994, *109* (2), 309–340.

Gomes, João F., "Financing Investment," *American Economic Review*, 2001, *91* (5), 1263–1285.

González, Beatriz, "Macroeconomics, Firm Dynamics and IPOs," Documentos de Trabajo 2030, Banco de España 2020.

Gopinath, Gita, Şebnem Kalemli-Özcan, Loukas Karabarbounis, and Carolina Villegas-Sanchez, "Capital Allocation and Productivity in South Europe," *Quarterly Journal of Economics*, 2017, *132* (4), 1915–1967.

Hadlock, Charles J. and Joshua R. Pierce, "New Evidence on Measuring Financial Constraints: Moving Beyond the KZ Index," *Review of Financial Studies*, 2010, *23* (5), 1909–1940.

Haltiwanger, John, Ron S. Jarmin, and Javier Miranda, "Who Creates Jobs? Small versus Large versus Young," *Review of Economics and Statistics*, 2013, *95* (2), 347–361.

Helwege, Jean and Nellie Liang, "Initial Public Offerings in Hot and Cold Markets," *Journal of Financial and Quantitative Analysis*, 2004, *39* (3), 541–569.

Hennessy, Christopher A. and Toni M. Whited, "Debt Dynamics," *Journal of Finance*, 2005, *60* (3), 1129–1165.

_ **and** _ , "How Costly Is External Financing? Evidence from a Structural Estimation," *Journal of Finance*, 2007, *62* (4), 1705–1745.

Hsieh, Chang-Tai and Peter J. Klenow, "Misallocation and Manufacturing TFP in China and India," *Quarterly Journal of Economics*, 2009, *124* (4), 1403–1448.

Ibbotson, Roger G. and Jeffrey F. Jaffe, ""Hot Issue" Markets," *Journal of Finance*, 1975, *30* (4), 1027–1042.

Ivanov, Vladimir and Craig M. Lewis, "The Determinants of Market-Wide Issue Cycles for Initial Public Offerings," *Journal of Corporate Finance*, 2008, *14* (5), 567–583.

Jermann, Urban and Vincenzo Quadrini, "Macroeconomic Effects of Financial Shocks," *American Economic Review*, 2012, *102* (1), 238–271.

Khan, Aubhik and Julia K. Thomas, "Credit Shocks and Aggregate Fluctuations in an Economy with Production Heterogeneity," *Journal of Political Economy*, 2013, *121* (6), 1055–1107.

Kiyotaki, Nobuhiro and John Moore, "Credit Cycles," *Journal of Political Economy*, 1997, *105* (2), 211–248.

Koijen, Ralph S. J. and Motohiro Yogo, "A Demand System Approach to Asset Pricing," *Journal of Political Economy*, 2019, *127* (4), 1475–1515.

Lowry, Michelle, "Why Does IPO Volume Fluctuate So Much?," *Journal of Financial Economics*, 2003, *67* (1), 3–40.

Midrigan, Virgiliu and Daniel Yi Xu, "Finance and Misallocation: Evidence from Plant-Level Data," *American Economic Review*, 2014, *104* (2), 422–458.

Moll, Benjamin, "Productivity Losses from Financial Frictions: Can Self-Financing Undo Capital Misallocation?," *American Economic Review*, 2014, *104* (10), 3186–3221.

Nola, Alessandro Di, "Capital Misallocation during the Great Recession," MPRA Paper 68289, University Library of Munich 2015.

Ottonello, Pablo and Thomas Winberry, "Financial Heterogeneity and the Investment Channel of Monetary Policy," *Econometrica*, 2020, *88* (6), 2473–2502.

Pagano, Marco, Fabio Panetta, and Luigi Zingales, "Why Do Companies Go Public? An Empirical Analysis," *Journal of Finance*, 1998, *53* (1), 27–64.

Pástor, Ľuboš and Pietro Veronesi, "Rational IPO Waves," *Journal of Finance*, 2005, *60* (4), 1713–1757.

Restuccia, Diego and Richard Rogerson, "Policy Distortions and Aggregate Productivity with Heterogeneous Establishments," *Review of Economic Dynamics*, 2008, *11* (4), 707–720.

Ritter, Jay R., "The "Hot Issue" Market of 1980," *Journal of Business*, 1984, *57* (2), 215–240.

Saunders, Anthony and Sascha Steffen, "The Costs of Being Private: Evidence from the Loan Market," *Review of Financial Studies*, 2011, *24* (12), 4091–4122.

Schenone, Carola, "Lending Relationships and Information Rents: Do Banks Exploit Their Information Advantages?," *Review of Financial Studies*, 2010, *23* (3), 1149–1199.

Sedláček, Petr and Vincent Sterk, "The Growth Potential of Startups over the Business Cycle," *American Economic Review*, 2017, *107* (10), 3182–3210.

Siemer, Michael, "Employment Effects of Financial Constraints during the Great Recession," *Review of Economics and Statistics*, 2019, *101* (1), 16–29.

Tran, Anh L. and Bang Nam Jeon, "The Dynamic Impact of Macroeconomic Factors on Initial Public Offerings: Evidence from Time-Series Analysis," *Applied Economics*, 2011, *43* (23), 3187–3201.

Yung, Chris, Gönül Çolak, and Wei Wang, "Cycles in the IPO Market," *Journal of Financial Economics*, 2008, *89* (1), 192–208.